Mathematical Statistics

Practice problems for the SNU Statistics graduate entrance exam

40 problems × 10 points. Solutions start on page 10.

Notation: $\mathrm{Exp}(\theta)$ has mean θ; $\mathrm{Gamma}(\alpha,\beta)$ has shape α and scale β; $\chi^2_\alpha(r)$, $t_\alpha(r)$, $F_\alpha(r_1,r_2)$, z_α are upper-α points. A sequence Y_n is $O_p(1)$ if for every $\epsilon>0$ there are M,N with $P(|Y_n|>M)<\epsilon$ for all $n\ge N$; $Y_n=o_p(1)$ means $Y_n \xrightarrow{p} 0$; $Y_n=O_p(a_n)$ means $Y_n/a_n=O_p(1)$.

Problems

Problem 1 *Analysis* [10 pts]

(a) Using the mean value theorem, show that $\dfrac{x}{1+x} < \log(1+x) < x$ for all $x>0$.

(b) Let $f_n(x)=nxe^{-nx}$ for $x\ge 0$. Find the pointwise limit f. Is the convergence uniform on $[0,\infty)$? On $[a,\infty)$ for a fixed $a>0$?

(c) Show that $\int_0^1 f_n(x)\,dx \to \int_0^1 f(x)\,dx$, even though the convergence in (b) is not uniform on $[0,1]$.

(d) Let $K\subset\mathbb{R}^k$ be compact and $g:K\to\mathbb{R}$ continuous. Prove that g is uniformly continuous on K.

Problem 2 *Joint, marginal and conditional distributions* [10 pts]

(X,Y) is uniformly distributed on $D=\{(x,y): 0<x<1,\ 0<y<x^2\}$.

(a) Find the joint pdf and the marginal pdfs of X and Y.

(b) Find $E(Y\mid X=x)$ and $\mathrm{Var}(Y\mid X=x)$, and use them to compute EY and $\mathrm{Var}\,Y$.

(c) Compute $\mathrm{Cov}(X,Y)$. Are X and Y independent?

(d) Compute $P(Y>X/2)$.

Problem 3 *Transformations* [10 pts]

Let $X,Y \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$.

(a) Find the joint pdf of $U=X/(X+Y)$ and $V=X+Y$. Show that they are independent and identify their distributions.

(b) Using (a), find $E(X\mid X+Y=v)$ and $E\big(X^2\mid X+Y=v\big)$.

(c) Find the pdf of $W=X-Y$.

(d) Let $m=\min(X,Y)$ and $R=\max(X,Y)-\min(X,Y)$. Show that m and R are independent and find their distributions.

Problem 4 *Multivariate normal* [10 pts]

Let $\mathbf{X}=(X_1,X_2,X_3)'\sim N_3(\mathbf{0},\Sigma)$ with $\Sigma=\begin{pmatrix}2&1&0\\1&2&1\\0&1&2\end{pmatrix}$.

(a) Find the distribution of $X_1 - X_2 + X_3$.

(b) Find the constant c for which $X_2 - c(X_1 + X_3)$ and $X_1 + X_3$ are independent, and justify independence.

(c) Find the conditional distribution of X_2 given $(X_1, X_3) = (x_1, x_3)$.

(d) Find the mgf $M(t) = E\exp(tX_1X_3)$ and state where it is finite.

(e) What is the distribution of $\mathbf{X}'\Sigma^{-1}\mathbf{X}$? Justify.

Problem 5 *Stochastic order and limits* **[10 pts]**

(a) Show that if $X_n \xrightarrow{d} X$, then $X_n = O_p(1)$.

(b) Show that if $X_n = O_p(1)$ and $Y_n = o_p(1)$, then $X_nY_n = o_p(1)$.

(c) Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$. Show that $n(\theta - X_{(n)}) \xrightarrow{d} \text{Exp}(\theta)$. Conclude that $X_{(n)} - \theta = O_p(n^{-1})$ and $\sqrt{n}(X_{(n)} - \theta) \xrightarrow{p} 0$.

(d) Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Without quoting the delta method, show $\sqrt{n}(\bar{X}_n^2 - \mu^2) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$, using the decomposition $\bar{X}_n^2 - \mu^2 = 2\mu(\bar{X}_n - \mu) + (\bar{X}_n - \mu)^2$.

Problem 6 *MLE, MME and efficiency* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from the Rayleigh density

$$f(x; \theta) = \frac{x}{\theta}\exp\Big(-\frac{x^2}{2\theta}\Big), \qquad x > 0,\ \theta > 0.$$

(a) Show that $X_1^2 \sim \text{Exp}(2\theta)$. Find the MLE $\hat{\theta}$, and its mean and variance.

(b) Compute the Fisher information $I(\theta)$ and show that $\hat{\theta}$ attains the Cramér–Rao lower bound. Is $\hat{\theta}$ the UMVUE?

(c) Using $EX_1 = \sqrt{\pi\theta/2}$, find the method-of-moments estimator $\tilde{\theta}$ based on $\bar{X}$ and its limiting distribution.

(d) Compute $ARE(\tilde{\theta}, \hat{\theta})$.

Problem 7 *Multinomial reparametrization* **[10 pts]**

A sample of n individuals is classified into three genotypes with counts

$$(N_1, N_2, N_3) \sim \text{Multi}\big(n;\ \theta^2,\ 2\theta(1-\theta),\ (1-\theta)^2\big), \qquad 0 < \theta < 1.$$

(a) Find the MLE $\hat{\theta}$.

(b) Show that $2N_1 + N_2 \sim B(2n, \theta)$, and find $E\hat{\theta}$ and $\text{Var}\,\hat{\theta}$.

(c) Find the Fisher information per individual and the limiting distribution of $\sqrt{n}(\hat{\theta} - \theta)$. Is $\hat{\theta}$ the UMVUE?

(d) With $n = 200$ and $(N_1, N_2, N_3) = (50, 90, 60)$, test at level 0.05 whether the three-cell model above fits, against a general trinomial, using Pearson's statistic. State the degrees of freedom.

Problem 8 *Sufficiency, completeness, UMVUE* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from $f(x;\theta) = e^{-(x-\theta)}$, $x > \theta$, $\theta \in \mathbb{R}$.

(a) Show that $X_{(1)}$ is sufficient, and find its distribution.

(b) Show that $X_{(1)}$ is complete.

(c) Find the UMVUEs of θ and of θ^2.

(d) Show that $X_{(1)}$ and $S = \sum_{i=1}^{n}(X_i - X_{(1)})$ are independent, and find the distribution of S.

Problem 9 *Most powerful and UMP tests* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from the Pareto density $f(x;\theta) = \theta x^{-\theta-1}$, $x > 1$, $\theta > 0$, and let $T = \sum_{i=1}^{n} \log X_i$.

(a) Show that $2\theta T \sim \chi^2(2n)$.

(b) Find the most powerful level-α test of $H_0 : \theta = 1$ vs $H_1 : \theta = \theta_1$ for a fixed $\theta_1 > 1$.

(c) Show that the test in (b) is UMP for $H_0 : \theta \le 1$ vs $H_1 : \theta > 1$, and express its power function with a chi-square cdf.

(d) For $n = 10$ and $\alpha = 0.05$ ($\chi^2_{0.95}(20) = 10.851$), give the rejection region and the power at $\theta = 2$.

Problem 10 *Likelihood ratio test* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta_1)$ and $Y_1, \dots, Y_m \overset{\text{iid}}{\sim} \text{Exp}(\theta_2)$ be independent samples. Consider $H_0 : \theta_1 = \theta_2$ vs $H_1 : \theta_1 \neq \theta_2$.

(a) Show that $2\log\Lambda$, the LRT statistic, is a function of $B = n\bar{X}/(n\bar{X} + m\bar{Y})$, and describe the form of the rejection region.

(b) Find the exact null distribution of B, and of $\bar{X}/\bar{Y}$.

(c) State the limiting null distribution of $2\log\Lambda$ as $n, m \to \infty$.

Problem 11 *Signed rank test* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from a continuous distribution symmetric about θ, and $W^+ = \sum_i \mathbf{1}\{X_i > 0\} R_i$, where R_i is the rank of $|X_i|$ among $|X_1|, \dots, |X_n|$. Under $H_0 : \theta = 0$:

(a) Show that $W^+ \overset{d}{=} \sum_{j=1}^{n} j B_j$ with $B_j \overset{\text{iid}}{\sim} \text{Bernoulli}(1/2)$. (You may use that the signs and the ranks of $|X_i|$ are independent.)

(b) Find EW^+ and $\operatorname{Var} W^+$, and the exact null distribution of W^+ for $n = 4$.

(c) Show that $W^+ = \sum_{1 \le i \le j \le n} \mathbf{1}\{X_i + X_j > 0\}$.

(d) The ARE of the signed rank test relative to the t test is $12\sigma^2\big(\int f^2\big)^2$, where f is the density of $X_1 - \theta$ and σ^2 its variance. Evaluate it for f uniform on $(-1, 1)$ and for $f(x) = \tfrac{1}{2}e^{-|x|}$.

Problem 12 *Bayes and minimax* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ with prior $\theta \sim \text{Exp}(\beta)$, i.e. Gamma$(1, \beta)$. Let $T = \sum X_i$ and $s = (n + 1/\beta)^{-1}$.

(a) Find the posterior distribution of θ.

(b) Find the Bayes estimator under squared error loss, and write it as a weighted average of $\bar{X}$ and the prior mean.

(c) Under the loss $L(\theta, a) = (a-\theta)^2/\theta$, show that the Bayes estimator is $a^* = sT$ and that its posterior expected loss is s for every T.

(d) Show that $\bar{X}$ is minimax for the loss in (c).

Problem 13 *Probability inequalities* **[10 pts]**

(a) Using Jensen's inequality, show that $(E|X|^r)^{1/r}$ is nondecreasing in $r > 0$.

(b) Let $\hat{p}$ be the proportion of successes in n Bernoulli(p) trials. Using Chebyshev's inequality, find n that guarantees $P(|\hat{p} - p| \geq 0.05) \leq 0.05$ for every p.

(c) (Chernoff bound) For $X \sim \text{Poisson}(\lambda)$, show $P(X \geq 2\lambda) \leq \inf_{t>0} e^{-2\lambda t} E e^{tX}$, and evaluate the bound.

Problem 14 *Conditional expectation and random sums* **[10 pts]**

Let $N \sim \text{Poisson}(\lambda)$ be independent of $X_1, X_2, \ldots$, which are i.i.d. with mean μ, variance σ^2 and mgf M_X. Let $S = \sum_{i=1}^{N} X_i$ ($S = 0$ if $N = 0$).

(a) Find ES and $\operatorname{Var} S$.

(b) Show that $M_S(t) = \exp\{\lambda(M_X(t) - 1)\}$.

(c) If $X_i \sim \text{Bernoulli}(p)$, identify the distribution of S.

(d) Show that $E(X_1 \mid S, N = n) = S/n$ for $n \geq 1$.

Problem 15 *Moment generating functions and cumulants* **[10 pts]**

(a) Identify the distribution with mgf $(0.3 + 0.7e^t)^5$.

(b) $X_1 \sim \text{Poisson}(2)$ and $X_2 \sim \text{Poisson}(1)$ are independent and $Y = X_1 + 2X_2$. Find the mgf, mean and variance of Y. Is Y Poisson?

(c) Show that the r-th cumulant of Gamma(α, β) is $\alpha\beta^r(r-1)!$, and find its skewness.

(d) A random variable satisfies $EX^k = (k+1)!$ for $k = 0, 1, 2, \ldots$. Identify its distribution.

Problem 16 *Uniform order statistics* **[10 pts]**

Let $U_{(1)} < \cdots < U_{(n)}$ be the order statistics of $n \geq 2$ i.i.d. $U(0,1)$ variables.

(a) Find the joint pdf of $(U_{(1)}, U_{(n)})$.

(b) Show that the range $R = U_{(n)} - U_{(1)}$ has the Beta($n-1, 2$) distribution, and find ER.

(c) Using (b), show that $\operatorname{Cov}(U_{(1)}, U_{(n)}) = \dfrac{1}{(n+1)^2(n+2)}$.

(d) Find $P(U_{(1)} < \frac{1}{2} < U_{(n)})$.

Problem 17 *Normal samples and prediction* **[10 pts]**

Let $X_1, \dots, X_{n+1} \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, and let $\bar{X}$ and S^2 be computed from the first n.

(a) Show that $\dfrac{X_{n+1} - \bar{X}}{S\sqrt{1 + 1/n}} \sim t(n-1)$, and give a 95% prediction interval for X_{n+1}.

(b) Show that $ES = \sigma\sqrt{\dfrac{2}{n-1}}\dfrac{\Gamma(n/2)}{\Gamma((n-1)/2)}$.

(c) Find c such that $c\sum_{i=1}^n (X_i - \bar{X})^2$ minimizes $E\big(c\sum(X_i - \bar{X})^2 - \sigma^2\big)^2$.

Problem 18 *Chi-square, t and F* **[10 pts]**

(a) If $T \sim t(r)$, show $T^2 \sim F(1, r)$.

(b) If $X \sim F(r_1, r_2)$, show $\dfrac{r_1 X}{r_2 + r_1 X} \sim \text{Beta}(r_1/2, r_2/2)$.

(c) $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ and $Y_1, \dots, Y_m \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ independently. Show $\bar{X}/\bar{Y} \sim F(2n, 2m)$.

(d) Show that $EX = r_2/(r_2 - 2)$ for $X \sim F(r_1, r_2)$, $r_2 > 2$.

Problem 19 *Bivariate normal* **[10 pts]**

(X, Y) is bivariate normal with means 0, variances 1 and correlation $\rho \in (-1, 1)$.

(a) Show that $P(X > 0, Y > 0) = \dfrac{1}{4} + \dfrac{\arcsin \rho}{2\pi}$.

(b) Show that $E\max(X, Y) = \sqrt{(1-\rho)/\pi}$.

(c) Find $E(Y^2 \mid X = x)$ and $\text{Cov}(X^2, Y^2)$.

Problem 20 *Quadratic forms* **[10 pts]**

Let $\mathbf{Z} \sim N_n(\mathbf{0}, I)$.

(a) For $n = 3$, show that $Q = \frac{1}{2}(Z_1 - Z_2)^2 + \frac{1}{6}(Z_1 + Z_2 - 2Z_3)^2 \sim \chi^2(2)$ and that $Q = \sum_{i=1}^3 (Z_i - \bar{Z})^2$.

(b) For symmetric A, show $E(\mathbf{Z}'A\mathbf{Z}) = \text{tr}\, A$ and $\text{Var}(\mathbf{Z}'A\mathbf{Z}) = 2\,\text{tr}(A^2)$.

(c) Let $J = \frac{1}{n}\mathbf{1}\mathbf{1}'$. Show that $\mathbf{Z}'J\mathbf{Z}$ and $\mathbf{Z}'(I - J)\mathbf{Z}$ are independent, and identify their distributions.

Problem 21 *Modes of convergence* **[10 pts]**

(a) Let $P(X_n = n) = 1/n$, $P(X_n = 0) = 1 - 1/n$. Show $X_n \xrightarrow{p} 0$ but $EX_n \not\to 0$.

(b) Let $Z \sim N(0, 1)$ and $X_n = (-1)^n Z$. Show $X_n \xrightarrow{d} Z$ but X_n does not converge in probability to Z.

(c) Show that if $X_n \xrightarrow{d} c$ for a constant c, then $X_n \xrightarrow{p} c$.

(d) Show that if $E(X_n - a)^2 \to 0$, then $X_n \xrightarrow{p} a$.

Problem 22 *Central limit theorem in practice* **[10 pts]**

(a) Approximate $P\big(\sum_{i=1}^{48} U_i > 26\big)$ for $U_i \overset{\text{iid}}{\sim} U(0,1)$.

(b) How large must n be so that a 95% interval $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ has half-width at most 0.03 regardless of p?

(c) Approximate $P(X \geq 120)$ for $X \sim \text{Poisson}(100)$ with a continuity correction.

Problem 23 *Delta method and variance stabilization* **[10 pts]**

(a) For a Poisson(θ) sample, find the limit law of $\sqrt{n}(\sqrt{\bar{X}} - \sqrt{\theta})$ and use it to build an asymptotic 95% interval for θ.

(b) For an Exp(θ) sample, find the limit law of $\sqrt{n}(\log \bar{X} - \log \theta)$.

(c) For $\hat{p} = X/n$, $X \sim B(n,p)$, find the limit law of $\sqrt{n}(\arcsin\sqrt{\hat{p}} - \arcsin\sqrt{p})$.

Problem 24 *Cauchy location* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from the Cauchy density $\frac{1}{\pi}\{1 + (x-\theta)^2\}^{-1}$.

(a) Using the characteristic function $Ee^{itX} = e^{i\theta t - |t|}$, show that $\bar{X}$ has the same distribution as X_1. Is $\bar{X}$ consistent?

(b) Find the limit law of $\sqrt{n}(\hat{m}_n - \theta)$, where $\hat{m}_n$ is the sample median.

(c) Compute the Fisher information $I(\theta)$ and the ARE of the median relative to an efficient estimator.

Problem 25 *A non-regular scale model* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from $U[\theta, 2\theta]$, $\theta > 0$.

(a) Find the MLE of θ and a minimal sufficient statistic.

(b) Find an unbiased estimator of θ that is a multiple of the MLE.

(c) Show that $(X_{(1)}, X_{(n)})$ is not complete.

Problem 26 *Two-parameter exponential* **[10 pts]**

Let $X_1, \dots, X_n$ ($n \geq 2$) be a sample from $f(x;\mu,\sigma) = \frac{1}{\sigma} e^{-(x-\mu)/\sigma}$, $x \geq \mu$.

(a) Find the MLEs $\hat{\mu}, \hat{\sigma}$.

(b) Show that $n(\hat{\mu} - \mu)/\sigma \sim \text{Exp}(1)$ and $2n\hat{\sigma}/\sigma \sim \chi^2(2n-2)$, independently.

(c) Given that $(X_{(1)}, \sum X_i)$ is complete sufficient, find the UMVUEs of σ and μ.

Problem 27 *Cramér–Rao bound* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ and $\eta = e^{-\theta} = P(X_1 = 0)$.

(a) Find the Cramér–Rao lower bound for unbiased estimators of η.

(b) The UMVUE of η is $(1-1/n)^T$, $T=\sum X_i$. Find its variance and show that it exceeds the bound for all $\theta>0$.

(c) Find the limit law of $\sqrt{n}(e^{-\bar X}-e^{-\theta})$ and compare.

Problem 28 *Normal with equal mean and variance* **[10 pts]**

Let $X_1,\dots,X_n$ be a sample from $N(\theta,\theta)$, $\theta>0$.

(a) Show that the MLE is $\hat\theta=\frac{1}{2}\big(-1+\sqrt{1+4m_2}\big)$, where $m_2=\frac{1}{n}\sum X_i^2$.

(b) Compute $I(\theta)$ and the limit law of $\sqrt{n}(\hat\theta-\theta)$.

(c) Compare $\hat\theta$ with $\bar X$ and S^2 by ARE.

Problem 29 *Kullback–Leibler divergence* **[10 pts]**

(a) Compute $KL(\theta_1,\theta_0)=E_{\theta_0}\log\{f(X;\theta_0)/f(X;\theta_1)\}$ for $N(\theta,1)$ and for Poisson(θ).

(b) Using $\log t\le t-1$, show $KL\ge 0$ with equality iff the densities coincide.

(c) Explain why this suggests that the MLE is consistent.

Problem 30 *Rao–Blackwellization* **[10 pts]**

(a) For a Poisson(θ) sample, compute $E\big(\mathbf{1}\{X_1=0\}\mid T\big)$, $T=\sum X_i$, and conclude.

(b) For a Bernoulli(θ) sample ($n\ge 2$), find the UMVUE of θ^2 by conditioning X_1X_2 on T.

(c) For $U(0,\theta)$, improve $2X_1$ by conditioning on $X_{(n)}$.

Problem 31 *Basu's theorem* **[10 pts]**

Let $X_1,\dots,X_n\overset{\text{iid}}{\sim}N(\mu,\sigma^2)$ with σ^2 known.

(a) Use Basu's theorem to show that $\bar X$ is independent of S^2 and of $X_{(n)}-X_{(1)}$.

(b) Show that $E\big(X_{(n)}\mid\bar X\big)=\bar X+c_n\sigma$ for a constant c_n not depending on μ.

(c) Find $E(X_1^2\mid\bar X)$.

Problem 32 *Tests for a normal variance* **[10 pts]**

Let $X_1,\dots,X_n\overset{\text{iid}}{\sim}N(\mu,\sigma^2)$.

(a) With μ known, find the UMP level-α test of $H_0:\sigma^2\le\sigma_0^2$ vs $H_1:\sigma^2>\sigma_0^2$, and its power function.

(b) With μ unknown, show that the LRT of $H_0:\sigma^2=\sigma_0^2$ vs $H_1:\sigma^2\ne\sigma_0^2$ is based on $v=\hat\sigma^2/\sigma_0^2$ through $n(v-1-\log v)$, and describe the rejection region.

Problem 33 *Power and sample size* **[10 pts]**

$X_1,\dots,X_n\overset{\text{iid}}{\sim}N(\mu,4^2)$; test $H_0:\mu\le 10$ vs $H_1:\mu>10$ at $\alpha=0.05$.

(a) Give the UMP test and its power function.

(b) Find the smallest n with power at least 0.9 at $\mu = 12$ ($z_{0.10} = 1.282$).

(c) With that n, compute the power at $\mu = 11$.

(d) Show that no UMP test exists for $H_0 : \mu = 10$ vs $H_1 : \mu \neq 10$.

Problem 34 *LRT for a uniform scale* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$; test $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.

(a) Find the LRT statistic and the size-α rejection region.

(b) Show that under H_0, $2 \log \Lambda \sim \chi^2(2)$ exactly. Why is this not $\chi^2(1)$?

Problem 35 *Wald, score and LRT* **[10 pts]**

$X \sim B(100, p)$ is observed as $x = 60$; test $H_0 : p = 0.5$ vs $H_1 : p \neq 0.5$ at level 0.05.

(a) Compute the LRT, Wald and score statistics.

(b) State the common limiting null distribution and the decisions.

(c) Which of the three needs only H_0, not the MLE? Give the corresponding 95% confidence set for p (the score interval) as an inequality.

Problem 36 *Pivots and confidence intervals* **[10 pts]**

(a) For an $\text{Exp}(\theta)$ sample, give an exact $1 - \alpha$ confidence interval for θ.

(b) For a $U(0, \theta)$ sample, show that among intervals $[X_{(n)}/b, X_{(n)}/a]$ with $0 < a < b \leq 1$ and coverage $1 - \alpha$, the shortest is $[X_{(n)}, X_{(n)}\alpha^{-1/n}]$.

(c) For $N(\mu, \sigma^2)$ with both unknown, give an exact interval for σ.

Problem 37 *Bayesian prediction and risk* **[10 pts]**

(a) Bernoulli(θ) trials with prior Beta(α, β) give t successes in n. Find $P(X_{n+1} = 1 \mid \mathbf{x})$. What is it for a uniform prior and $t = n$?

(b) $X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$, σ^2 known, prior $N(\mu_0, \tau^2)$. Show that the posterior mean has Bayes risk $(n/\sigma^2 + 1/\tau^2)^{-1}$, smaller than the Bayes risk σ^2/n of $\bar{X}$.

(c) In (b), find the frequentist MSE of the posterior mean as a function of θ, and show that it exceeds σ^2/n when $|\theta - \mu_0|$ is large.

Problem 38 *Wilcoxon rank sum* **[10 pts]**

Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent samples from continuous distributions, $N = m + n$, and $W = \sum_j R(Y_j)$, the sum of the ranks of the Y's in the pooled sample.

(a) Under H_0 (same distribution), show that every rank R has $ER = \frac{N+1}{2}$, $\text{Var}\, R = \frac{N^2-1}{12}$, and that two distinct ranks have covariance $-\frac{N+1}{12}$.

(b) Deduce $EW = \frac{n(N+1)}{2}$ and $\text{Var}\, W = \frac{mn(N+1)}{12}$.

(c) For $m = n = 3$, find $P(W \geq 14)$ under H_0.

Problem 39 *Analysis: limits and series* **[10 pts]**

(a) Using Problem 1(a), prove $\lim_{n\to\infty}(1 + x/n)^n = e^x$ for $x > 0$.

(b) Show that $\sum_{k=0}^{\infty} x^k/k!$ converges uniformly on $[-a, a]$ for every $a > 0$.

(c) Justify differentiating this series term by term, and conclude that its sum $E(x)$ satisfies $E' = E$.

(d) Show that a function with $f'' \geq 0$ on an interval is convex, using the mean value theorem.

Problem 40 *Linear algebra* **[10 pts]**

(a) Show that a symmetric idempotent matrix has eigenvalues 0 or 1 and that its rank equals its trace.

(b) For X of size $n \times p$ with rank p, show that $H = X(X'X)^{-1}X'$ and $I - H$ are symmetric idempotent with traces p and $n - p$, and that $(I - H)X = 0$.

(c) Prove $(I + \mathbf{a}\mathbf{b}')^{-1} = I - \dfrac{\mathbf{a}\mathbf{b}'}{1 + \mathbf{b}'\mathbf{a}}$ when $1 + \mathbf{b}'\mathbf{a} \neq 0$, and $\det(I + \mathbf{a}\mathbf{b}') = 1 + \mathbf{b}'\mathbf{a}$.

(d) Use (c) to invert $\Sigma = (1 - \rho)I_k + \rho\mathbf{1}\mathbf{1}'$, and find for which ρ it is positive definite.

Solutions

Solution 1

(a) Apply the MVT to $\log(1+t)$ on $[0,x]$: $\log(1+x) = \frac{x}{1+\xi}$ for some $\xi \in (0,x)$. Since $1 < 1+\xi < 1+x$, we get $\frac{x}{1+x} < \log(1+x) < x$.

(b) $f_n(0) = 0$, and for $x > 0$, $nxe^{-nx} \to 0$ because $ue^{-u} \to 0$ as $u \to \infty$. So $f \equiv 0$. Since $\frac{d}{dx} f_n = ne^{-nx}(1-nx)$, f_n peaks at $x = 1/n$ with $\sup_{x\ge 0} f_n = f_n(1/n) = e^{-1} \not\to 0$: not uniform on $[0,\infty)$. On $[a,\infty)$, for $n > 1/a$ the function is decreasing there, so $\sup_{x\ge a} f_n = nae^{-na} \to 0$: uniform.

(c) $\int_0^1 nxe^{-nx}dx = \frac{1}{n}\int_0^n ue^{-u}du \le \frac{1}{n}\int_0^\infty ue^{-u}du = \frac{1}{n} \to 0 = \int_0^1 f$.

(d) Suppose not. Then there are $\epsilon > 0$ and $x_n, y_n \in K$ with $|x_n - y_n| < 1/n$ but $|g(x_n) - g(y_n)| \ge \epsilon$. By compactness (Bolzano–Weierstrass), a subsequence $x_{n_k} \to x \in K$, and then $y_{n_k} \to x$ as well. By continuity $g(x_{n_k}) - g(y_{n_k}) \to g(x) - g(x) = 0$, contradicting $\ge \epsilon$.

Solution 2

(a) The area of D is $\int_0^1 x^2 dx = \frac{1}{3}$, so $f(x,y) = 3$ on D. Then $f_X(x) = \int_0^{x^2} 3\,dy = 3x^2$ on $(0,1)$, and $f_Y(y) = \int_{\sqrt{y}}^1 3\,dx = 3(1-\sqrt{y})$ on $(0,1)$.

(b) Given $X = x$, $Y \sim U(0,x^2)$: $E(Y \mid x) = x^2/2$, $\operatorname{Var}(Y \mid x) = x^4/12$. Then $EY = E(X^2)/2 = \frac{1}{2}\cdot\frac{3}{5} = \frac{3}{10}$, and

$$\operatorname{Var} Y = E\Big(\frac{X^4}{12}\Big) + \operatorname{Var}\Big(\frac{X^2}{2}\Big) = \frac{1}{12}\cdot\frac{3}{7} + \frac{1}{4}\Big(\frac{3}{7} - \frac{9}{25}\Big) = \frac{1}{28} + \frac{3}{175} = \frac{37}{700}.$$

(c) $E(XY) = E\{X\,E(Y \mid X)\} = E(X^3)/2 = \frac{1}{2}\cdot\frac{3}{6} = \frac{1}{4}$, $EX = \frac{3}{4}$, so $\operatorname{Cov}(X,Y) = \frac{1}{4} - \frac{3}{4}\cdot\frac{3}{10} = \frac{1}{40}$. Not independent (nonzero covariance; also the support is not a product set).

(d) $\{y > x/2\} \cap D$ is nonempty only where $x^2 > x/2$, i.e. $x > 1/2$:

$$P(Y > X/2) = \int_{1/2}^1 3\Big(x^2 - \frac{x}{2}\Big)dx = 3\Big[\frac{x^3}{3} - \frac{x^2}{4}\Big]_{1/2}^1 = 3\Big(\frac{1}{12} + \frac{1}{48}\Big) = \frac{5}{16}.$$

Solution 3

(a) Inverse map $x = uv$, $y = (1-u)v$, $0 < u < 1$, $v > 0$, with $|J| = |\det \begin{pmatrix} v & u \\ -v & 1-u \end{pmatrix}| = v$. So $f(u,v) = e^{-v}v = \mathbf{1}_{(0,1)}(u)\cdot ve^{-v}$: $U \sim U(0,1)$ and $V \sim \text{Gamma}(2,1)$, independent.

(b) $X = UV$ with U independent of V, so $E(X \mid V = v) = vEU = v/2$ and $E(X^2 \mid V = v) = v^2 EU^2 = v^2/3$.

(c) For $w > 0$, $f_W(w) = \int_0^\infty f_X(y+w)f_Y(y)\,dy = \int_0^\infty e^{-(y+w)}e^{-y}dy = \frac{1}{2}e^{-w}$; by symmetry $f_W(w) = \frac{1}{2}e^{-|w|}$, $w \in \mathbb{R}$ (double exponential).

(d) The order statistics have joint pdf $2e^{-y_1-y_2}$ on $0 < y_1 < y_2$. With $m = y_1$, $r = y_2 - y_1$ ($|J| = 1$), $f(m,r) = 2e^{-2m-r} = 2e^{-2m}\cdot e^{-r}$ on $m, r > 0$. So $m \sim \text{Exp}(1/2)$ and $R \sim \text{Exp}(1)$ are independent (memorylessness).

Solution 4

(a) With $a = (1,-1,1)'$, $\Sigma a = (1,0,1)'$ and $a'\Sigma a = 2$. So $X_1 - X_2 + X_3 \sim N(0,2)$.

(b) $(X_2 - c(X_1+X_3),\ X_1+X_3)$ is a linear image of $\mathbf{X}$, hence bivariate normal, and is independent iff the covariance is 0. $\operatorname{Cov}(X_2, X_1+X_3) = 1+1 = 2$ and $\operatorname{Var}(X_1+X_3) = 2+2+2\cdot 0 = 4$, so $\operatorname{Cov} = 2 - 4c = 0$ gives $c = \frac{1}{2}$.

(c) $\Sigma_{12} = (1,1)$, $\Sigma_{22} = 2I_2$, so $\Sigma_{12}\Sigma_{22}^{-1} = (\frac{1}{2},\frac{1}{2})$. Hence $X_2 \mid (x_1,x_3) \sim N\big(\frac{x_1+x_3}{2},\ 2 - \frac{1}{2}(1+1)\big) = N\big(\frac{x_1+x_3}{2}, 1\big)$.

(d) $\mathrm{Cov}(X_1, X_3) = 0$ and they are jointly normal, so $X_1, X_3 \overset{\text{iid}}{\sim} N(0,2)$. Conditioning on X_3, $E(e^{tX_1X_3} \mid X_3) = \exp(\frac{1}{2} \cdot 2t^2X_3^2) = e^{t^2X_3^2}$. Since $X_3^2/2 \sim \chi^2(1)$, $Ee^{t^2X_3^2} = Ee^{2t^2(X_3^2/2)} = (1-4t^2)^{-1/2}$. So $M(t) = (1-4t^2)^{-1/2}$ for $|t| < \frac{1}{2}$, and $M(t) = \infty$ otherwise.

(e) Σ is positive definite (leading minors $2, 3, 4$). $\mathbf{Z} = \Sigma^{-1/2}\mathbf{X} \sim N_3(\mathbf{0}, I)$ and $\mathbf{X}'\Sigma^{-1}\mathbf{X} = \mathbf{Z}'\mathbf{Z} \sim \chi^2(3)$.

Solution 5

(a) Let F be the cdf of X and $\epsilon > 0$. Since F has at most countably many discontinuities, choose M such that $\pm M$ are continuity points and $P(X > M) + P(X \le -M) < \epsilon/2$. Then

$$P(|X_n| > M) \le 1 - F_n(M) + F_n(-M) \to 1 - F(M) + F(-M) < \epsilon/2,$$

so $P(|X_n| > M) < \epsilon$ for all large n; the finitely many earlier n are handled by enlarging M.

(b) For $\delta, \epsilon > 0$, choose M with $P(|X_n| > M) < \epsilon$ for large n. Then

$$P(|X_nY_n| > \delta) \le P(|X_n| > M) + P(|Y_n| > \delta/M) < \epsilon + o(1).$$

(c) For $0 \le x \le n\theta$, $P(n(\theta - X_{(n)}) > x) = P(X_{(n)} < \theta - x/n) = (1 - \frac{x}{n\theta})^n \to e^{-x/\theta}$. So $n(\theta - X_{(n)}) \xrightarrow{d} \mathrm{Exp}(\theta)$, which is $O_p(1)$ by (a), i.e. $X_{(n)} - \theta = O_p(n^{-1})$. Then $\sqrt{n}(X_{(n)} - \theta) = n^{-1/2} \cdot n(X_{(n)} - \theta) = o(1)O_p(1) = o_p(1)$ by (b).

(d) By the CLT, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$, so $2\mu\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$. Also $\sqrt{n}(\bar{X}_n - \mu)^2 = \sqrt{n}(\bar{X}_n - \mu) \cdot (\bar{X}_n - \mu) = O_p(1)\, o_p(1) = o_p(1)$, by (a), the WLLN and (b). Slutsky's theorem gives the result.

Solution 6

(a) $P(X^2 > t) = \int_{\sqrt{t}}^{\infty} \frac{x}{\theta} e^{-x^2/2\theta} dx = e^{-t/(2\theta)}$, so $X^2 \sim \mathrm{Exp}(2\theta)$. The log-likelihood is $l(\theta) = \sum \log x_i - n \log\theta - \sum x_i^2/(2\theta)$, with $\dot{l} = -n/\theta + \sum x_i^2/(2\theta^2)$ positive iff $\theta < \sum x_i^2/(2n)$. So $\hat{\theta} = \frac{1}{2n}\sum X_i^2$, with $E\hat{\theta} = \theta$ and $\mathrm{Var}\,\hat{\theta} = \frac{(2\theta)^2}{4n} = \frac{\theta^2}{n}$.

(b) $\partial_\theta^2 \log f = \frac{1}{\theta^2} - \frac{x^2}{\theta^3}$, so $I(\theta) = -\frac{1}{\theta^2} + \frac{2\theta}{\theta^3} = \frac{1}{\theta^2}$. The CR bound for unbiased estimators is $\frac{1}{nI(\theta)} = \frac{\theta^2}{n} = \mathrm{Var}\,\hat{\theta}$. So $\hat{\theta}$ is efficient, hence UMVUE (also: $\sum X_i^2$ is complete sufficient in this exponential family).

(c) $\theta = \frac{2}{\pi}(EX)^2$, so $\tilde{\theta} = \frac{2}{\pi}\bar{X}^2$. $\mathrm{Var}\,X = EX^2 - (EX)^2 = 2\theta - \frac{\pi\theta}{2} = \frac{(4-\pi)\theta}{2}$. With $g(m) = 2m^2/\pi$, $g'(m) = 4m/\pi$ and $m = \sqrt{\pi\theta/2}$, the delta method gives

$$\sqrt{n}(\tilde{\theta} - \theta) \xrightarrow{d} N\Big(0, \frac{16m^2}{\pi^2} \cdot \frac{(4-\pi)\theta}{2}\Big) = N\Big(0, \frac{4(4-\pi)}{\pi}\theta^2\Big).$$

(d) $ARE(\tilde{\theta}, \hat{\theta}) = \dfrac{\theta^2}{4(4-\pi)\theta^2/\pi} = \dfrac{\pi}{4(4-\pi)} \approx 0.915.$

Solution 7

(a) $l(\theta) = (2N_1 + N_2)\log\theta + (N_2 + 2N_3)\log(1-\theta) + \text{const}$, strictly concave, so $\hat{\theta} = \frac{2N_1+N_2}{2n}$.

(b) Each individual contributes $Y_i \in \{2, 1, 0\}$ copies of the allele, with probabilities $\theta^2, 2\theta(1-\theta), (1-\theta)^2$, i.e. $Y_i \sim B(2, \theta)$, independently. So $2N_1 + N_2 = \sum Y_i \sim B(2n, \theta)$, $E\hat{\theta} = \theta$, $\mathrm{Var}\,\hat{\theta} = \frac{\theta(1-\theta)}{2n}$.

(c) Per individual, $I(\theta) = \mathrm{Var}\left(\frac{Y}{\theta} - \frac{2-Y}{1-\theta}\right) = \frac{2\theta(1-\theta)}{\theta^2(1-\theta)^2} = \frac{2}{\theta(1-\theta)}$. Hence $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta(1-\theta)/2)$ (here even $\mathrm{Var}\,\hat{\theta} = 1/(nI)$ exactly). $2N_1 + N_2$ is complete sufficient (binomial family), and $\hat{\theta}$ is an unbiased function of it: UMVUE.

(d) $\hat{\theta} = (100 + 90)/400 = 0.475$. Expected counts: $200(0.475)^2 = 45.125$, $200 \cdot 2(0.475)(0.525) = 99.75$, $200(0.525)^2 = 55.125$. Pearson's statistic:

$$\frac{4.875^2}{45.125} + \frac{9.75^2}{99.75} + \frac{4.875^2}{55.125} = 0.527 + 0.953 + 0.431 = 1.91.$$

Degrees of freedom $= \dim\Omega - \dim\Omega_0 = 2-1 = 1$; since $1.91 < \chi^2_{0.05}(1) = 3.84$, the model is not rejected.

Solution 8

(a) $\prod f(x_i;\theta) = e^{-\sum x_i + n\theta}\mathbf{1}\{x_{(1)} > \theta\} = \big[e^{n\theta}\mathbf{1}\{x_{(1)} > \theta\}\big]e^{-\sum x_i}$; by factorization $X_{(1)}$ is sufficient. $P(X_{(1)} > y) = e^{-n(y-\theta)}$ for $y > \theta$, so $X_{(1)} - \theta \sim \mathrm{Exp}(1/n)$.

(b) If $E_\theta g(X_{(1)}) = \int_\theta^\infty g(y) n e^{-n(y-\theta)}dy = 0$ for all θ, then $\int_\theta^\infty g(y)e^{-ny}dy = 0$ for all θ. Differentiating in θ, $g(\theta)e^{-n\theta} = 0$ for almost every θ, so $g = 0$ a.e. and $P_\theta(g(X_{(1)}) = 0) = 1$.

(c) $EX_{(1)} = \theta + \frac{1}{n}$, so $X_{(1)} - \frac{1}{n}$ is the UMVUE of θ (Lehmann–Scheffé). Also $EX^2_{(1)} = \mathrm{Var}\, X_{(1)} + (EX_{(1)})^2 = \frac{1}{n^2} + \theta^2 + \frac{2\theta}{n} + \frac{1}{n^2}$, so

$$E\Big[X^2_{(1)} - \frac{2}{n}X_{(1)}\Big] = \theta^2 + \frac{2\theta}{n} + \frac{2}{n^2} - \frac{2\theta}{n} - \frac{2}{n^2} = \theta^2,$$

and $X^2_{(1)} - \frac{2}{n}X_{(1)}$ is the UMVUE of θ^2.

(d) Write $X_i = \theta + Z_i$ with $Z_i \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$; then $S = \sum(Z_i - Z_{(1)})$ does not depend on θ: it is ancillary. $X_{(1)}$ is complete sufficient, so by Basu's theorem they are independent. With $D_i = (n-i+1)(Z_{(i)} - Z_{(i-1)}) \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$ (exponential spacings), $S = \sum_{i=2}^n (Z_{(i)} - Z_{(1)}) = \sum_{i=2}^n D_i \sim \mathrm{Gamma}(n-1, 1)$.

Solution 9

(a) $P(\log X > t) = P(X > e^t) = e^{-\theta t}$, so $\log X_i \overset{\text{iid}}{\sim} \mathrm{Exp}(1/\theta)$, $T \sim \mathrm{Gamma}(n, 1/\theta)$ and $2\theta T \sim \mathrm{Gamma}(n, 2) = \chi^2(2n)$.

(b) $\dfrac{\prod f(x_i;\theta_1)}{\prod f(x_i;1)} = \theta_1^n \exp\{-(\theta_1 - 1)T\}$, decreasing in T for $\theta_1 > 1$. By Neyman–Pearson, the MP test rejects for $T \le c$. Under $\theta = 1$, $2T \sim \chi^2(2n)$, so $c = \frac{1}{2}\chi^2_{1-\alpha}(2n)$: reject when $2T \le \chi^2_{1-\alpha}(2n)$.

(c) The test does not depend on θ_1, so it is UMP for $\theta = 1$ vs $\theta > 1$. Its power is

$$\gamma(\theta) = P_\theta\big(2T \le \chi^2_{1-\alpha}(2n)\big) = P\big(\chi^2(2n) \le \theta\chi^2_{1-\alpha}(2n)\big),$$

increasing in θ. So $\max_{\theta\le 1}\gamma(\theta) = \gamma(1) = \alpha$, and any level-$\alpha$ test of $\theta \le 1$ is level α for $\theta = 1$; hence the test is UMP for $H_0 : \theta \le 1$. (This is also the one-parameter exponential family result with $g(\theta) = -\theta$ decreasing in θ, applied to T.)

(d) Reject when $2T \le 10.851$, i.e. $\sum \log X_i \le 5.43$. Power at $\theta = 2$: $P(\chi^2(20) \le 21.70) \approx 0.64$.

Solution 10

(a) Unrestricted MLEs: $\hat\theta_1 = \bar X$, $\hat\theta_2 = \bar Y$; under H_0: $\hat\theta_0 = (n\bar X + m\bar Y)/(n+m)$. Since $l(\theta_1,\theta_2) = -n\log\theta_1 - n\bar x/\theta_1 - m\log\theta_2 - m\bar y/\theta_2$, the terms $n\bar x/\hat\theta$ etc. sum to $n+m$ in both cases, and

$$2\log\Lambda = 2\big\{(n+m)\log\hat\theta_0 - n\log\bar X - m\log\bar Y\big\}.$$

With $B = n\bar X/(n\bar X + m\bar Y)$, $\bar X = \frac{(n+m)\hat\theta_0}{n}B$ and $\bar Y = \frac{(n+m)\hat\theta_0}{m}(1-B)$, so

$$2\log\Lambda = -2\{n\log B + m\log(1-B)\} + 2\{n\log\tfrac{n}{n+m} + m\log\tfrac{m}{n+m}\}.$$

The function $-n\log b - m\log(1-b)$ is convex with minimum at $b = n/(n+m)$, so the LRT rejects when $B \le c_1$ or $B \ge c_2$, where $c_1 < \frac{n}{n+m} < c_2$ give equal values of the statistic and total size α.

(b) Under H_0 (common θ), $n\bar X/\theta \sim \mathrm{Gamma}(n,1)$ and $m\bar Y/\theta \sim \mathrm{Gamma}(m,1)$ are independent, so $B \sim \mathrm{Beta}(n,m)$, free of θ. Equivalently $\frac{2n\bar X/\theta}{2n}\Big/\frac{2m\bar Y/\theta}{2m} = \bar X/\bar Y \sim F(2n,2m)$.

(c) $\dim\Omega - \dim\Omega_0 = 2-1 = 1$, so $2\log\Lambda \xrightarrow{d} \chi^2(1)$ under H_0.

Solution 11

(a) Let D_j be the index of the observation whose $|X|$ has rank j. Then $W^+ = \sum_{j=1}^{n} j\,\mathbf{1}\{X_{D_j} > 0\}$. Since the sign vector is independent of the ranks and its entries are i.i.d. fair, the signs attached to ranks $1, \dots, n$ are again i.i.d. fair, so $W^+ \overset{d}{=} \sum j B_j$.

(b) $EW^+ = \frac{1}{2}\sum j = \frac{n(n+1)}{4}$ and $\operatorname{Var} W^+ = \frac{1}{4}\sum j^2 = \frac{n(n+1)(2n+1)}{24}$. For $n = 4$, W^+ is the sum of a uniformly random subset of $\{1, 2, 3, 4\}$ (16 subsets):

w	0	1	2	3	4	5	6	7	8	9	10
$16\,P(W^+ = w)$	1	1	1	2	2	2	2	2	1	1	1

(c) Order by absolute value: $|Y_{(1)}| < \cdots < |Y_{(n)}|$. Then $W^+ = \sum_i i\mathbf{1}\{Y_{(i)} > 0\} = \sum_i \mathbf{1}\{Y_{(i)} > 0\} + \sum_{j<i} \mathbf{1}\{Y_{(i)} > 0\}$, and for $j < i$, $Y_{(i)} > 0 \iff Y_{(i)} + Y_{(j)} > 0$ since $|Y_{(j)}| < |Y_{(i)}|$. Hence $W^+ = \sum_{i \le j} \mathbf{1}\{X_i + X_j > 0\}$.

(d) Uniform on $(-1, 1)$: $\int f^2 = \frac{1}{2}$, $\sigma^2 = \frac{1}{3}$, so $ARE = 12 \cdot \frac{1}{3} \cdot \frac{1}{4} = 1$. Double exponential: $\int f^2 = \int \frac{1}{4}e^{-2|x|} = \frac{1}{4}$, $\sigma^2 = 2$, so $ARE = 12 \cdot 2 \cdot \frac{1}{16} = \frac{3}{2}$.

Solution 12

(a) $\pi(\theta \mid \mathbf{x}) \propto e^{-n\theta}\theta^T \cdot e^{-\theta/\beta} = \theta^T e^{-\theta/s}$, so $\theta \mid \mathbf{x} \sim \text{Gamma}(T + 1, s)$.

(b) The posterior mean is $s(T + 1) = \dfrac{n}{n + 1/\beta}\bar{X} + \dfrac{1/\beta}{n + 1/\beta}\beta$, a weighted average of $\bar{X}$ and the prior mean β.

(c) The posterior expected loss is $\rho(a) = a^2 E(\theta^{-1} \mid \mathbf{x}) - 2a + E(\theta \mid \mathbf{x})$. For $T \ge 1$, $E(\theta^{-1} \mid \mathbf{x}) = \frac{1}{sT}$, so ρ is minimized at $a^* = sT$ with $\rho(a^*) = s(T + 1) - sT = s$. For $T = 0$, $E(\theta^{-1} \mid \mathbf{x}) = \infty$, so only $a = 0 = sT$ has finite loss, $\rho(0) = E(\theta \mid \mathbf{x}) = s$. In all cases the Bayes estimator is sT and its posterior expected loss is s, so its Bayes risk is $r(\pi_\beta) = s = (n + 1/\beta)^{-1}$.

(d) $\bar{X}$ has risk $E\big[(\bar{X} - \theta)^2/\theta\big] = \frac{\theta/n}{\theta} = \frac{1}{n}$ for all θ. For any estimator δ, $\max_\theta R_\delta(\theta) \ge r(\pi_\beta, \delta) \ge (n + 1/\beta)^{-1}$ for every β; letting $\beta \to \infty$, $\max_\theta R_\delta \ge \frac{1}{n} = \max_\theta R_{\bar{X}}$. So $\bar{X}$ is minimax (limit of Bayes estimators, least favourable sequence of priors).

Solution 13

(a) For $0 < r < s$, $u \mapsto u^{s/r}$ is convex on $[0, \infty)$, so by Jensen $E|X|^s = E\{(|X|^r)^{s/r}\} \ge (E|X|^r)^{s/r}$. Take the $1/s$ power.

(b) $\operatorname{Var}\hat{p} = p(1 - p)/n \le 1/(4n)$, so $P(|\hat{p} - p| \ge 0.05) \le \frac{1}{4n(0.05)^2} = \frac{100}{n} \le 0.05$ once $n \ge 2000$.

(c) For $t > 0$, Markov's inequality applied to e^{tX} gives $P(X \ge 2\lambda) \le e^{-2\lambda t} E e^{tX} = \exp\{-2\lambda t + \lambda(e^t - 1)\}$. The exponent is minimized at $e^t = 2$, giving $P(X \ge 2\lambda) \le e^{\lambda(1 - 2\log 2)} = (e/4)^\lambda$.

Solution 14

(a) $E(S \mid N) = N\mu$ and $\operatorname{Var}(S \mid N) = N\sigma^2$, so $ES = \lambda\mu$ and $\operatorname{Var} S = E(N\sigma^2) + \operatorname{Var}(N\mu) = \lambda(\sigma^2 + \mu^2)$.

(b) $E(e^{tS} \mid N) = M_X(t)^N$, and $Es^N = e^{\lambda(s-1)}$ for Poisson N; so $M_S(t) = \exp\{\lambda(M_X(t) - 1)\}$.

(c) $M_X(t) = 1 - p + pe^t$ gives $M_S(t) = \exp\{\lambda p(e^t - 1)\}$: $S \sim \text{Poisson}(\lambda p)$ (thinning).

(d) Given $N = n$, $(X_1, \dots, X_n)$ is exchangeable, so $E(X_i \mid S, N = n)$ is the same for all $i \le n$; these sum to $E(S \mid S, N = n) = S$. Hence each equals S/n.

Solution 15

(a) $B(5, 0.7)$.

(b) $M_Y(t) = M_{X_1}(t)M_{X_2}(2t) = \exp\{2(e^t - 1) + (e^{2t} - 1)\}$, $EY = 2 + 2 \cdot 1 = 4$, $\operatorname{Var} Y = 2 + 4 \cdot 1 = 6$. Not Poisson, since its mean and variance differ.

(c) $K(t) = -\alpha\log(1-\beta t) = \alpha\sum_{r\ge1}\frac{(\beta t)^r}{r}$, so $\kappa_r = r!\cdot\frac{\alpha\beta^r}{r} = \alpha\beta^r(r-1)!$. Skewness $= \kappa_3/\kappa_2^{3/2} = 2\alpha\beta^3/(\alpha\beta^2)^{3/2} = 2/\sqrt{\alpha}$.

(d) $M(t) = \sum_k EX^k\frac{t^k}{k!} = \sum_k (k+1)t^k = (1-t)^{-2}$ for $|t|<1$. This is the Gamma$(2,1)$ mgf, and an mgf finite near 0 determines the distribution.

Solution 16

(a) $f(u,v) = n(n-1)(v-u)^{n-2}$ on $0<u<v<1$.

(b) With $r = v-u$ and $u\in(0,1-r)$, $f_R(r) = \int_0^{1-r} n(n-1)r^{n-2}du = n(n-1)r^{n-2}(1-r)$, the Beta$(n-1,2)$ density $(1/B(n-1,2) = n(n-1))$. $ER = \frac{n-1}{n+1}$.

(c) $U_{(1)}\sim$ Beta$(1,n)$ and $U_{(n)}\sim$ Beta$(n,1)$ both have variance $\frac{n}{(n+1)^2(n+2)}$, and $\operatorname{Var} R = \frac{2(n-1)}{(n+1)^2(n+2)}$. From $\operatorname{Var} R = \operatorname{Var} U_{(n)} + \operatorname{Var} U_{(1)} - 2\operatorname{Cov}$,

$$\operatorname{Cov}(U_{(1)},U_{(n)}) = \frac{2n-2(n-1)}{2(n+1)^2(n+2)} = \frac{1}{(n+1)^2(n+2)}.$$

(d) $1 - P(\text{all} < \frac12) - P(\text{all} > \frac12) = 1 - 2^{1-n}$.

Solution 17

(a) $X_{n+1} - \bar X \sim N(0,\sigma^2(1+1/n))$, and it is independent of S^2 (X_{n+1} is independent of the first n, and $\bar X$ is independent of S^2). Dividing by $S\sqrt{1+1/n}$ gives $Z/\sqrt{V/(n-1)}$ with $V = (n-1)S^2/\sigma^2\sim\chi^2(n-1)$: a $t(n-1)$ variable. Prediction interval: $\bar X \pm t_{0.025}(n-1)S\sqrt{1+1/n}$.

(b) With $V\sim\chi^2(n-1)$,

$$E\sqrt{V} = \int_0^\infty \frac{v^{n/2-1}e^{-v/2}}{\Gamma(\frac{n-1}{2})2^{(n-1)/2}}dv = \frac{\Gamma(\frac n2)2^{n/2}}{\Gamma(\frac{n-1}{2})2^{(n-1)/2}} = \sqrt2\frac{\Gamma(n/2)}{\Gamma((n-1)/2)},$$

and $ES = \frac{\sigma}{\sqrt{n-1}}E\sqrt{V}$.

(c) With $\sum(X_i-\bar X)^2 = \sigma^2V$, $EV = n-1$, $EV^2 = 2(n-1)+(n-1)^2$,

$$E(c\sigma^2V-\sigma^2)^2 = \sigma^4\{c^2(n-1)(n+1) - 2c(n-1)+1\},$$

minimized at $c = \frac{1}{n+1}$.

Solution 18

(a) $T^2 = Z^2/(V/r)$ with $Z^2\sim\chi^2(1)$ independent of $V\sim\chi^2(r)$: this is $F(1,r)$.

(b) Write $X = \frac{V_1/r_1}{V_2/r_2}$. Then $\frac{r_1X}{r_2+r_1X} = \frac{V_1}{V_1+V_2}$ with independent $V_i\sim$ Gamma$(r_i/2,2)$, which is Beta$(r_1/2,r_2/2)$.

(c) $2n\bar X/\theta\sim\chi^2(2n)$ and $2m\bar Y/\theta\sim\chi^2(2m)$ are independent, and $\bar X/\bar Y = \frac{(2n\bar X/\theta)/(2n)}{(2m\bar Y/\theta)/(2m)}\sim F(2n,2m)$.

(d) $EX = \frac{r_2}{r_1}EV_1\,EV_2^{-1} = \frac{r_2}{r_1}\cdot r_1\cdot\frac{1}{r_2-2}$, since $EV^{-1} = \frac{\Gamma(r/2-1)2^{r/2-1}}{\Gamma(r/2)2^{r/2}} = \frac{1}{r-2}$ for $V\sim\chi^2(r)$.

Solution 19

(a) Write $Y = \rho X + \sqrt{1-\rho^2}Z$ with X,Z i.i.d. $N(0,1)$, and $X = R\cos\Theta$, $Z = R\sin\Theta$ with $\Theta\sim U(-\pi,\pi)$ independent of R. With $\phi = \arcsin\rho$, $Y = R\sin(\Theta+\phi)$. So $\{X>0, Y>0\} = \{\Theta\in(-\frac\pi2,\frac\pi2)\cap(-\phi,\pi-\phi)\} = \{\Theta\in(-\phi,\frac\pi2)\}$, of probability $\frac{\pi/2+\phi}{2\pi} = \frac14 + \frac{\arcsin\rho}{2\pi}$.

(b) $\max(X,Y) = \frac{X+Y}{2} + \frac{|X-Y|}{2}$, and $X-Y\sim N(0,2(1-\rho))$ with $E|X-Y| = \sqrt{2(1-\rho)}\sqrt{2/\pi}$. Hence $E\max(X,Y) = \sqrt{(1-\rho)/\pi}$.

(c) $Y\mid X=x\sim N(\rho x, 1-\rho^2)$, so $E(Y^2\mid x) = \rho^2x^2+1-\rho^2$. Then $EX^2Y^2 = E\{X^2(\rho^2X^2+1-\rho^2)\} = 3\rho^2+1-\rho^2$ and $\operatorname{Cov}(X^2,Y^2) = 2\rho^2$.

Solution 20

(a) $\mathbf{a} = (1,-1,0)'/\sqrt{2}$ and $\mathbf{b} = (1,1,-2)'/\sqrt{6}$ are orthonormal, so $U = \mathbf{a}'\mathbf{Z}$, $V = \mathbf{b}'\mathbf{Z}$ are i.i.d. $N(0,1)$ and $Q = U^2 + V^2 \sim \chi^2(2)$. Together with $\mathbf{1}/\sqrt{3}$ they form an orthonormal basis, so $\sum Z_i^2 = 3\bar{Z}^2 + U^2 + V^2$, i.e. $Q = \sum Z_i^2 - 3\bar{Z}^2 = \sum (Z_i - \bar{Z})^2$.

(b) Write $A = P\Lambda P'$ with P orthogonal and $\mathbf{Y} = P'\mathbf{Z} \sim N(\mathbf{0}, I)$. Then $\mathbf{Z}'A\mathbf{Z} = \sum \lambda_i Y_i^2$, with mean $\sum \lambda_i = \operatorname{tr} A$ and variance $\sum \lambda_i^2 \operatorname{Var}(Y_i^2) = 2\sum \lambda_i^2 = 2 \operatorname{tr} A^2$.

(c) J and $I - J$ are symmetric idempotent and $J(I-J) = 0$, so $J\mathbf{Z}$ and $(I-J)\mathbf{Z}$ are uncorrelated jointly normal vectors, hence independent; the quadratic forms are their squared lengths. Since $\operatorname{tr} J = 1$ and $\operatorname{tr}(I-J) = n-1$, they are $\chi^2(1)$ and $\chi^2(n-1)$.

Solution 21

(a) $P(|X_n| > \epsilon) = 1/n \to 0$, but $EX_n = n \cdot \frac{1}{n} = 1$ for all n.

(b) By symmetry, $X_n \sim N(0,1)$ for every n, so $X_n \xrightarrow{d} Z$ trivially. For odd n, $X_n - Z = -2Z$ and $P(|2Z| > \epsilon)$ is a fixed positive number, so $X_n \not\xrightarrow{p} Z$.

(c) For $\epsilon > 0$, $c - \epsilon$ and $c + \epsilon/2$ are continuity points of the limit cdf $\mathbf{1}\{x \geq c\}$, so

$$P(|X_n - c| \geq \epsilon) \leq P(X_n \leq c - \epsilon) + 1 - P(X_n \leq c + \epsilon/2) \to 0 + 0.$$

(d) Markov's inequality: $P(|X_n - a| \geq \epsilon) \leq E(X_n - a)^2/\epsilon^2 \to 0$.

Solution 22

(a) The sum has mean 24 and variance $48/12 = 4$, so $P \approx 1 - \Phi\big(\frac{26-24}{2}\big) = 1 - \Phi(1) = 0.159$.

(b) Since $p(1-p) \leq \frac{1}{4}$, we need $1.96\sqrt{0.25/n} \leq 0.03$, i.e. $n \geq (1.96/0.06)^2 = 1067.1$, so $n = 1068$.

(c) $P(X \geq 120) = P(X > 119.5) \approx 1 - \Phi\big(\frac{19.5}{10}\big) = 1 - \Phi(1.95) = 0.026$.

Solution 23

(a) $g(\theta) = \sqrt{\theta}$, $g'(\theta) = \frac{1}{2\sqrt{\theta}}$, and $\sqrt{n}(\bar{X} - \theta) \xrightarrow{d} N(0,\theta)$, so $\sqrt{n}(\sqrt{\bar{X}} - \sqrt{\theta}) \xrightarrow{d} N(0, \frac{1}{4})$, free of θ. Hence $\sqrt{\theta} \in \sqrt{\bar{X}} \pm \frac{1.96}{2\sqrt{n}}$, and squaring (lower end truncated at 0) gives the interval for θ.

(b) $g = \log$, $g'(\theta) = 1/\theta$, variance $\theta^2 \cdot \theta^{-2} = 1$: the limit is $N(0,1)$.

(c) $g(p) = \arcsin\sqrt{p}$ has $g'(p) = \frac{1}{2\sqrt{p(1-p)}}$, so the limit is $N\big(0, \frac{p(1-p)}{4p(1-p)}\big) = N(0, \frac{1}{4})$.

Solution 24

(a) $Ee^{it\bar{X}} = \{\varphi(t/n)\}^n = e^{i\theta t - |t|}$, so $\bar{X} \overset{d}{=} X_1$. Then $P(|\bar{X} - \theta| > \epsilon) = P(|X_1 - \theta| > \epsilon)$ does not tend to 0: $\bar{X}$ is not consistent.

(b) The median of X_1 is θ with $f(\theta) = 1/\pi$, so $\sqrt{n}(\hat{m}_n - \theta) \xrightarrow{d} N(0, \frac{1}{4f(\theta)^2}) = N(0, \frac{\pi^2}{4})$.

(c) The score is $\frac{2(x-\theta)}{1+(x-\theta)^2}$, so with Y standard Cauchy,

$$I(\theta) = E\frac{4Y^2}{(1+Y^2)^2} = \frac{4}{\pi}\int_{-\infty}^{\infty} \frac{y^2}{(1+y^2)^3} dy = \frac{4}{\pi} \cdot \frac{\pi}{8} = \frac{1}{2},$$

using $y = \tan u$: $\int \sin^2 u \cos^2 u \, du = \frac{\pi}{8}$ over $(-\frac{\pi}{2}, \frac{\pi}{2})$. The efficient limiting variance is $1/I = 2$, so $ARE(\text{median}, \text{MLE}) = \frac{2}{\pi^2/4} = \frac{8}{\pi^2} \approx 0.81$.

Solution 25

(a) $L(\theta) = \theta^{-n}\mathbf{1}\{x_{(n)}/2 \leq \theta \leq x_{(1)}\}$, decreasing on its support, so $\hat{\theta} = X_{(n)}/2$. L depends on the data only through $(x_{(1)}, x_{(n)})$, and two samples give proportional likelihoods only if these agree, so $(X_{(1)}, X_{(n)})$ is minimal sufficient.

(b) $X_i = \theta(1+U_i)$ with $U_i \overset{\text{iid}}{\sim} U(0,1)$, so $EX_{(n)} = \theta\frac{2n+1}{n+1}$ and $EX_{(1)} = \theta\frac{n+2}{n+1}$. Thus $\frac{n+1}{2n+1}X_{(n)} = \frac{2(n+1)}{2n+1}\hat\theta$ is unbiased.

(c) $g = (2n+1)X_{(1)} - (n+2)X_{(n)}$ has $E_\theta g = \frac{\theta}{n+1}\{(2n+1)(n+2) - (n+2)(2n+1)\} = 0$ for all θ, but $P(g=0) = 0$. So the statistic is not complete.

Solution 26

(a) $L = \sigma^{-n}\exp\{-\sum(x_i-\mu)/\sigma\}\mathbf{1}\{\mu \le x_{(1)}\}$ increases in μ up to $x_{(1)}$, so $\hat\mu = X_{(1)}$. Then $\log L = -n\log\sigma - \sum(x_i - x_{(1)})/\sigma$ is maximized at $\hat\sigma = \bar X - X_{(1)}$.

(b) $X_i = \mu + \sigma Z_i$ with $Z_i \overset{\text{iid}}{\sim} \text{Exp}(1)$. The normalized spacings $D_i = (n-i+1)(Z_{(i)} - Z_{(i-1)})$ are i.i.d. Exp(1). So $n(\hat\mu-\mu)/\sigma = D_1 \sim \text{Exp}(1)$, and $n\hat\sigma/\sigma = \sum(Z_i - Z_{(1)}) = \sum_{i\ge 2} D_i \sim \text{Gamma}(n-1,1)$, i.e. $2n\hat\sigma/\sigma \sim \chi^2(2n-2)$, independent of D_1.

(c) $E(n\hat\sigma) = (n-1)\sigma$, so $\tilde\sigma = \frac{n\hat\sigma}{n-1} = \frac{1}{n-1}\sum(X_i - X_{(1)})$ is the UMVUE of σ. Since $EX_{(1)} = \mu + \sigma/n$, the UMVUE of μ is $X_{(1)} - \tilde\sigma/n$.

Solution 27

(a) $I(\theta) = 1/\theta$, so the bound is $\frac{(de^{-\theta}/d\theta)^2}{nI(\theta)} = \frac{\theta e^{-2\theta}}{n}$.

(b) $Es^T = e^{n\theta(s-1)}$, so $E(1-\frac{1}{n})^{2T} = \exp\{n\theta((1-\frac{1}{n})^2 - 1)\} = e^{-2\theta+\theta/n}$, and

$$\text{Var}\left((1-\tfrac{1}{n})^T\right) = e^{-2\theta}(e^{\theta/n} - 1) > e^{-2\theta}\frac{\theta}{n},$$

since $e^x - 1 > x$ for $x > 0$. No unbiased estimator attains the bound.

(c) By the delta method, $\sqrt{n}(e^{-\bar X} - e^{-\theta}) \xrightarrow{d} N(0, \theta e^{-2\theta})$. Also $n\,\text{Var}\left((1-\frac{1}{n})^T\right) = e^{-2\theta}n(e^{\theta/n}-1) \to \theta e^{-2\theta}$: the UMVUE and the MLE are asymptotically equivalent and asymptotically attain the bound.

Solution 28

(a) $l(\theta) = -\frac{n}{2}\log\theta - \frac{1}{2\theta}\sum(x_i-\theta)^2 + c$, and

$$\dot l(\theta) = -\frac{n}{2\theta} + \frac{\sum x_i^2}{2\theta^2} - \frac{n}{2} = \frac{n}{2\theta^2}(m_2 - \theta - \theta^2).$$

It is positive for $\theta < \hat\theta$ and negative after, where $\hat\theta$ is the positive root of $\theta^2 + \theta - m_2 = 0$.

(b) $I(\theta) = \frac{1}{\theta} + \frac{1}{2\theta^2} = \frac{2\theta+1}{2\theta^2}$ (mean part plus variance part of a normal), so $\sqrt{n}(\hat\theta - \theta) \xrightarrow{d} N\big(0, \frac{2\theta^2}{2\theta+1}\big)$. (Check by the delta method: $\text{Var}\,X^2 = 2\theta^2 + 4\theta^3$ and $g'(m_2) = 1/(2\theta+1)$.)

(c) $\bar X$ has variance θ/n and S^2 asymptotic variance $2\theta^2/n$. So $ARE(\bar X, \hat\theta) = \frac{2\theta}{2\theta+1}$ and $ARE(S^2, \hat\theta) = \frac{1}{2\theta+1}$, both below 1.

Solution 29

(a) Normal: $E_{\theta_0}\frac{1}{2}\{(X-\theta_1)^2 - (X-\theta_0)^2\} = \frac{1}{2}(\theta_1-\theta_0)^2$. Poisson: $\log\frac{f_0}{f_1} = x\log\frac{\theta_0}{\theta_1} - \theta_0 + \theta_1$, so $KL = \theta_0\log\frac{\theta_0}{\theta_1} - \theta_0 + \theta_1$.

(b) $-KL = E_0\log\frac{f_1}{f_0} \le E_0\big(\frac{f_1}{f_0} - 1\big) = \int_{f_0>0} f_1 - 1 \le 0$, with equality iff $f_1/f_0 = 1$ with probability one.

(c) By the LLN, $\frac{1}{n}l_n(\theta) \to E_{\theta_0}\log f(X;\theta)$ for each θ, and by (b) this limit is uniquely maximized at θ_0. The maximizer of $\frac{1}{n}l_n$, the MLE, should then approach θ_0 (rigorous under uniform convergence or concavity).

Solution 30

(a) $X_1 \mid T = t \sim B(t, 1/n)$, so $E(\mathbf{1}\{X_1 = 0\} \mid T) = (1-1/n)^T$, the UMVUE of $e^{-\theta}$ (T is complete sufficient).

(b) $E(X_1X_2 \mid T=t) = P(X_1=X_2=1 \mid T=t) = \binom{n-2}{t-2}/\binom{n}{t} = \frac{t(t-1)}{n(n-1)}$; the UMVUE of θ^2 is $\frac{T(T-1)}{n(n-1)}$.

(c) Given $X_{(n)}=y$, $X_1=y$ with probability $\frac{1}{n}$ and otherwise is $U(0,y)$. So $E(2X_1 \mid X_{(n)}) = 2X_{(n)}\big(\frac{1}{n}+\frac{n-1}{2n}\big) = \frac{n+1}{n}X_{(n)}$.

Solution 31

(a) With σ^2 known, $\bar X$ is complete sufficient for μ. S^2 and $X_{(n)}-X_{(1)}$ do not change when every X_i is shifted, so their laws do not depend on μ: they are ancillary, and Basu's theorem gives independence.

(b) $X_{(n)}-\bar X$ is also location invariant, hence independent of $\bar X$. So $E(X_{(n)} \mid \bar X) = \bar X + E(X_{(n)} - \bar X) = \bar X + \sigma E(Z_{(n)})$ with $Z_i \overset{\text{iid}}{\sim} N(0,1)$: $c_n = EZ_{(n)}$.

(c) $X_1-\bar X \sim N(0,\sigma^2\frac{n-1}{n})$ is independent of $\bar X$, so $E(X_1^2 \mid \bar X) = \bar X^2 + \sigma^2\frac{n-1}{n}$.

Solution 32

(a) The density is $\exp\{-T/(2\sigma^2) - \frac{n}{2}\log(2\pi\sigma^2)\}$ in $T=\sum(x_i-\mu)^2$, with $-1/(2\sigma^2)$ increasing in σ^2. The UMP test rejects for large T: $T \ge \sigma_0^2\chi^2_\alpha(n)$, since $T/\sigma_0^2 \sim \chi^2(n)$ at σ_0^2. Power: $\gamma(\sigma^2) = P\big(\chi^2(n) \ge \sigma_0^2\chi^2_\alpha(n)/\sigma^2\big)$.

(b) With $\hat\sigma^2 = \frac{1}{n}\sum(x_i-\bar x)^2$, $l(\sigma^2) = -\frac{n}{2}\log\sigma^2 - \frac{n\hat\sigma^2}{2\sigma^2}$ at $\mu=\bar x$, so

$$2\{l(\hat\sigma^2) - l(\sigma_0^2)\} = n\Big(\frac{\hat\sigma^2}{\sigma_0^2} - 1 - \log\frac{\hat\sigma^2}{\sigma_0^2}\Big) = n(v-1-\log v).$$

This is convex in v with minimum at $v=1$, so we reject for $v \le c_1$ or $v \ge c_2$, where $nv = (n-1)S^2/\sigma_0^2 \sim \chi^2(n-1)$ under H_0.

Solution 33

(a) Reject when $\bar X \ge 10 + 1.645\cdot 4/\sqrt{n}$; the power is $\gamma(\mu) = \Phi\big(\sqrt{n}(\mu-10)/4 - 1.645\big)$.

(b) $\sqrt{n}\cdot\frac{2}{4} - 1.645 \ge 1.282$ gives $n \ge \{2(1.645+1.282)\}^2 = 34.3$: $n=35$.

(c) $\gamma(11) = \Phi(\sqrt{35}/4 - 1.645) = \Phi(-0.166) = 0.434$.

(d) The MP test against $\mu_1 > 10$ rejects for large $\bar X$, and against $\mu_1 < 10$ for small $\bar X$ (Neyman–Pearson, with uniqueness of MP tests). No single test is MP against both.

Solution 34

(a) $L(\theta) = \theta^{-n}\mathbf{1}\{\theta \ge x_{(n)}\}$ and $\hat\theta = X_{(n)}$, so $\Lambda = L(\hat\theta)/L(\theta_0) = (\theta_0/X_{(n)})^n$ if $X_{(n)} \le \theta_0$ and ∞ otherwise. We reject when $X_{(n)} > \theta_0$ or $X_{(n)} \le c$; the size is $P_{\theta_0}(X_{(n)} \le c) = (c/\theta_0)^n = \alpha$, so $c = \theta_0\alpha^{1/n}$.

(b) Under H_0, $U = (X_{(n)}/\theta_0)^n \sim U(0,1)$, and $2\log\Lambda = -2\log U \sim \text{Exp}(2) = \chi^2(2)$. The usual $\chi^2(1)$ limit requires the regularity conditions; here the support depends on θ, and the MLE converges at rate n rather than $\sqrt{n}$.

Solution 35

(a) $\hat p = 0.6$.

- LRT: $2\{60\log 1.2 + 40\log 0.8\} = 2(10.939 - 8.926) = 4.03$.
- Wald: $\frac{(0.6-0.5)^2}{0.6\cdot 0.4/100} = 4.17$.
- Score: $\frac{(60-50)^2}{100\cdot 0.25} = 4.00$.

(b) Each is asymptotically $\chi^2(1)$ under H_0. All exceed 3.84, so H_0 is rejected (the score statistic only barely).

(c) The score statistic uses only p_0. The score confidence set is $\{p : (60-100p)^2 \le 3.84\cdot 100p(1-p)\}$, an interval found by solving the quadratic.

Solution 36

(a) $2n\bar{X}/\theta \sim \chi^2(2n)$ gives $\Big[\dfrac{2n\bar{X}}{\chi^2_{\alpha/2}(2n)}, \dfrac{2n\bar{X}}{\chi^2_{1-\alpha/2}(2n)}\Big]$.

(b) $W = X_{(n)}/\theta$ has cdf w^n, so the coverage is $P(a \le W \le b) = b^n - a^n = 1 - \alpha$, and the length is $X_{(n)}(1/a - 1/b)$. Treating $a = a(b)$ through $a^n = b^n - (1-\alpha)$, $a'(b) = b^{n-1}/a^{n-1}$, and

$$\frac{d}{db}\Big(\frac{1}{a} - \frac{1}{b}\Big) = -\frac{b^{n-1}}{a^{n+1}} + \frac{1}{b^2} < 0,$$

because $a < b$. The length decreases in b, so $b = 1$ and $a = \alpha^{1/n}$ are optimal.

(c) $\Big[\sqrt{(n-1)S^2/\chi^2_{\alpha/2}(n-1)},\ \sqrt{(n-1)S^2/\chi^2_{1-\alpha/2}(n-1)}\Big]$.

Solution 37

(a) $P(X_{n+1} = 1 \mid \mathbf{x}) = E(\theta \mid \mathbf{x}) = \frac{t+\alpha}{n+\alpha+\beta}$; with a uniform prior and $t = n$ this is $\frac{n+1}{n+2}$ (Laplace's rule of succession).

(b) The posterior variance $v = (n/\sigma^2 + 1/\tau^2)^{-1}$ does not depend on $\mathbf{x}$, so the Bayes risk of the posterior mean is $E\{\mathrm{Var}(\theta \mid \mathbf{X})\} = v < \sigma^2/n$. The Bayes risk of $\bar{X}$ is its constant MSE σ^2/n.

(c) The posterior mean is $w\bar{X} + (1-w)\mu_0$ with $w = \frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2} \in (0,1)$. Its MSE is $w^2\frac{\sigma^2}{n} + (1-w)^2(\theta-\mu_0)^2$, which exceeds $\frac{\sigma^2}{n}$ iff $(\theta-\mu_0)^2 > \frac{(1+w)\sigma^2}{(1-w)n}$.

Solution 38

(a) Under H_0 the rank vector is uniform over permutations, so each rank is uniform on $\{1,\dots,N\}$: mean $\frac{N+1}{2}$, variance $\frac{N^2-1}{12}$. All N ranks sum to a constant, so $0 = N\,\mathrm{Var}\,R + N(N-1)c$, giving $c = -\frac{N+1}{12}$.

(b) $EW = \frac{n(N+1)}{2}$ and

$$\mathrm{Var}\,W = n\frac{N^2-1}{12} - n(n-1)\frac{N+1}{12} = \frac{n(N+1)(N-n)}{12} = \frac{mn(N+1)}{12}.$$

(c) The $\binom{6}{3} = 20$ subsets of $\{1,\dots,6\}$ are equally likely; sums ≥ 14 come only from $\{4,5,6\}$ and $\{3,5,6\}$. So $P(W \ge 14) = 2/20 = 0.1$.

Solution 39

(a) Problem 1(a) with x/n gives $\frac{nx}{n+x} < n\log(1+x/n) < x$. Both bounds tend to x, and exp is continuous.

(b) $|x^k/k!| \le a^k/k!$ on $[-a,a]$, and $\sum a^k/k! < \infty$ by the ratio test; apply the Weierstrass M-test.

(c) The series of derivatives $\sum_{k\ge1} x^{k-1}/(k-1)!$ is the same series, uniformly convergent on $[-a,a]$, and the original converges at 0. By the theorem on term-by-term differentiation, $E'(x) = \sum_{k\ge1} x^{k-1}/(k-1)! = E(x)$.

(d) By the MVT, $f'' \ge 0$ makes f' nondecreasing. For $x < y < z$, the MVT gives $\frac{f(y)-f(x)}{y-x} = f'(\xi_1) \le f'(\xi_2) = \frac{f(z)-f(y)}{z-y}$, with $\xi_1 < y < \xi_2$. With $y = tx + (1-t)z$, this rearranges to $f(y) \le tf(x) + (1-t)f(z)$.

Solution 40

(a) If $P\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \ne \mathbf{0}$, then $\lambda\mathbf{v} = P^2\mathbf{v} = \lambda^2\mathbf{v}$, so $\lambda \in \{0,1\}$. A symmetric P is orthogonally diagonalizable, so its rank is the number of eigenvalues equal to 1, which is their sum, $\mathrm{tr}\,P$.

(b) $H' = H$, $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$, and $\mathrm{tr}\,H = \mathrm{tr}\{(X'X)^{-1}X'X\} = p$. The same holds for $I - H$, with trace $n - p$, and $(I-H)X = X - X = 0$.

(c) $(I+\mathbf{a}\mathbf{b}')\big(I-\frac{\mathbf{a}\mathbf{b}'}{1+\mathbf{b}'\mathbf{a}}\big)=I+\mathbf{a}\mathbf{b}'\big(1-\frac{1+\mathbf{b}'\mathbf{a}}{1+\mathbf{b}'\mathbf{a}}\big)=I$. The matrix $\mathbf{a}\mathbf{b}'$ has eigenvalue $\mathbf{b}'\mathbf{a}$ (eigenvector $\mathbf{a}$) and 0 with multiplicity $k-1$, so $\det(I+\mathbf{a}\mathbf{b}')=1+\mathbf{b}'\mathbf{a}$.

(d) For $\rho\neq 1$, $\Sigma=(1-\rho)\big(I+\frac{\rho}{1-\rho}\mathbf{1}\mathbf{1}'\big)$, so

$$\Sigma^{-1}=\frac{1}{1-\rho}\Big(I-\frac{\rho}{1+(k-1)\rho}\mathbf{1}\mathbf{1}'\Big).$$

The eigenvalues of Σ are $1+(k-1)\rho$ (eigenvector $\mathbf{1}$) and $1-\rho$ (multiplicity $k-1$), so Σ is positive definite iff $-\frac{1}{k-1}<\rho<1$.